DISCOVER_PI_RULE

RESULTS: DISCOVER_PI_RULE

PREREG sha256 f836369c6672361504992d3fe47f988c5a6f2e1a9701b51d699921f6773b96ee, hashed before the tests. Script hashes are in scripts/SHA256SUMS.

Verdict: OBSTRUCTION. Residual Category A structure does not force a unique non-degenerate 539-step Π word. RP1 shows that the word is exactly the seed mod 3^539, and every word occurs. RP2 shows that P_res only marks the local pattern “01”.

Test	Result	Category
T1: C10 on Π*_C1 = 0^539 (forced by C1, trivial-seed word)	1 segment → **FAIL**	A
T2: C10 on Π*_max = (01)^269 0 (sha256 c2c26fda…)	539 segments → **FAIL**	A (exact); the rule itself is a free extremal choice
T3c: RP4 uniqueness depends on the w_539 convention	810 maximisers without it → **PASS** (confirms the uniqueness is fragile)	A
Map-C6	**NOT RUN** (mod-4 S-profile undefined in the source)	—

Map-C10: every canonical candidate fails. In fact, any word with exactly 4 constant-branch segments has at most 3 “01” transitions, so it carries at most 3 P_res hits (Category A). C10 is therefore essentially incompatible with a resonant (hit-rich) Π. Map-C6 stays OPEN/undefined. **Bridge: NO.**

Open next: the smallest forcing pattern would be a source-written selection on x mod 3^539 (for example “Π := the max-P_res word with terminal hit”, which would make RP4 the rule), or an operational definition of C6’s “mod-4 profile”. Both are decisions for Arvin.

NOTES: DISCOVER_PI_RULE (exploration first, then proofs)

Inventory of residual objects that could index a length-539 sequence

O_res = {21+9j : 0≤j<539} (539 elements, 8 β♯ blocks 3×68 + 5×67); β♯/β♮; τ and g = (−1)^τ (accepted); C1 “index rounds by O_res”; C3/Principle (S) 18+521; residue-word bijection; MF2/DM1–DM5 hit law; Lemma V; P_res (accepted).

Proofs (Category A)

- RP1: each T3 branch k ↦ k, 4k+2, 2k+1 is injective in k, and w_0 = x mod 3. By induction, w_0..w_{n−1} is determined by x mod 3^n and determines it. Checked exhaustively for n ≤ 8.

- RP2: u ≡ 3 (9) ⇔ u = 3k with k ≡ 1 (3) ⇔ w_i = 0 and T3(u) = k ≡ 1 (3), i.e. w_{i+1} = 1. Checked for u < 10^6.
- RP3: T3(21+9j) = 7+3j ≡ 1 (mod 3), and every element of O_res is ≡ 0 (mod 3). So no T3 step stays in O_res, and the "carrier walk" is empty.
- RP4: "01" occurrences cannot overlap, so a 540-letter word has at most 270 of them, with equality only for (01)^270. Confirmed by DP. Without w_539: the maximum is 269, reached by 810 words (tests T3c).
- MF2/DM laws and Lemma V are statements about u mod 2^k. By CRT, the seed's 2-adic part is independent of x mod 3^539 (RP1). They are distributional, not word-selecting: they cannot pick a single word. (Category A for the independence; the laws themselves come from the prior packages.)
- 18+521: nothing in the sources gives a branch letter to either block. Splitting a word 18+521 needs a letter assignment, which would be a free choice.
- β♯ schedule as word: the sources contain no forced map {0..7} → {0,1,2}. Dropped as an invention.

Remaining free choices (any one of these would be needed to fix Π)

1. A seed, or a selection rule on x mod 3^539 (RP1: all 3^539 words are realised).
2. An extremal objective (for example max P_res hits) plus the endpoint convention for w_539.
3. An alphabet map from fibre index or block to branch letter.

PREREG: DISCOVER_PI_RULE

Written 2026-10-24 ~9:05 PM CDT after exploration (scripts/explore.py, data/explore.json, hash data/explore.sha256), before tests. Owner: Arvin Hampton. New files only. CORE_FREEZE. Residual-vs-Resonant lock. No C11 re-run. Labels RP/*Π* only.

Exploration facts used (Category A, proofs in NOTES.md)

- RP1 (word = seed mod 3^n): the T3 branch word of length n is a bijection of x mod 3^n onto {0,1,2}^n. Every word is realised; seeds form one residue class mod 3^539.
- RP2 (P_res is local): u ≡ 3 (9) ⇔ w_i = 0 and w_{i+1} = 1. P_res constrains words only through the 2-letter pattern "01".
- RP3 (carrier walk empty): T3(O_res) ∩ O_res = ∅, since T3(21+9j) = 7+3j ≡ 1 (3).
- RP4 (extremal word): over w_0..w_539 (hits at steps 0..538), the maximum hit count is 270, reached by exactly one word, (01)^270. Its 539-step word is Π*_max = (01)^269 0.
- C1 literal: σ(i) = sorted O_res[i] is forced and unique, but its branch word is Π*_C1 = 0^539. That is the trivial-seed word x = 3^539 z.

Candidate rules (neither is in a written source)

- Π*_C1: forced by C1 alone. Not a T3 trajectory through O_res (RP3).
- Π*_max: forced only after choosing the objective "maximise P_res hits", plus the convention that w_539 counts. That is an extremal choice, so it is a free choice.

Tests (deterministic, exact)

- T1 (Map-C10 on Π*_C1): the prediction is exactly 4 maximal constant-branch segments with lengths ordered like 1/34 < 5/2 < 35/4. Expected: FAIL (1 segment).

- T2 (Map-C10 on Π*_max): same prediction. Expected: FAIL (539 segments).
- T3c (robustness): count the words with the maximum hit count when hits are restricted to steps 0..537 (no w_539). Prediction: more than 1, i.e. the RP4 uniqueness depends on that convention.
- Map-C6: NOT RUN. “Word read mod 4 has M_S-order profile” has no operational definition in the source, and making one up would be an invention. Verdict rule: a unique Π is “found” only if a rule needing no free choice yields a non-degenerate T3 word. Otherwise the verdict is OBSTRUCTION. Bridge: NO in all cases.